Revision history

The model and text were revised after rounds of adversarial review of the code and the draft. Each entry lists what changed. The current code is published as thirlwall_statistical_model.py, with technical notes and a model specification, and reproduces every figure in the article. Entries below use the working names of the time, sim.py to sim5.py. The published file began as sim5.py renamed and documented, with two speed changes (optional Numba compilation, and detection checks limited to the twelve months they use) and extra printed lines; later corrections are listed below.

Revised 2 October 2026 after peer review of the code. Rolling rules are now checked from the first monitoring month; baselines update as time passes; alarm episodes are counted alongside alarm unit-years; the rota review takes its deaths from the alarm that triggered it and reports three tests separately; detection is run for each kind of unit; the base rate is widened to 1–30 per 10,000; and the count of nurses in the frame is made conditional on q. The headline numbers moved. The conclusions did not.

Revised again, 2 October 2026, after a second review. Pooled rota figures are now weighted by the number of units of each kind; the correct rota test is calculated exactly from each roster; rosters persist across alarms, so distinct nurses are counted; the offender is placed on the rota, so the chance that a flagged nurse is the offender is computed rather than inferred; a perfectly risk-adjusted CUSUM chart is added as a benchmark; the link between the recommendations is stated conditionally; and the robustness check is included in the published code. Again, the numbers moved slightly and the conclusions held.

Third revision, 2 October 2026. Rule E is now described as an idealised benchmark whose false-alarm rate is a setting, with a threshold sweep showing the trade-off; the offender effect is labelled as expected deaths, with an exact-count robustness check; Recommendation 10 is acknowledged; detection runs use 10,000 simulations per cell; the CUSUM review window covers the whole excursion; and the posterior uses the simulated flag rate directly. The code is sim4.py.

Fourth revision, 3 October 2026. The idealised chart is now calibrated on crossings per unit-year, using one fixed set of 50,000 null paths per unit type for every threshold, so the stated false-alarm rates mean what they say: closely at the E10 and E50 settings, approximately at the strictest settings, where the calibration rests on as few as 50 simulated crossings. The threshold sweep is extended to 1 false alarm per 10,000 unit-years; it shows no floor, and the earlier claim of one is withdrawn. Background detection, with no offender present, is now reported, and the correct rota test is described as conservative under a discrete null. The code is sim5.py.

Fifth revision, 3 October 2026. Wording only, after a final review: the false-to-true ratio is stated in unit-years throughout; the extreme end of the threshold sweep is described as approximate; the correct rota test is qualified as correct under this model; and the posterior's event definitions are disclosed. No figures changed.

Sixth revision, 3 October 2026. Structure only: a short summary and a map of the sections now open the piece, a wrap-up closes the argument before the limits, and this history moved to its own tab. No figures changed.

Seventh revision, 3 October 2026. Consistency check against the published code. "About 17 distinct nurses" is corrected to 16 (17 is the number of flagging episodes); the top of the chance-alarm range is corrected from 33 to 34; the code now prints the two figures it previously did not (the medians in the step-two table, and rule D's alarm rate in local units); and the three rota tests are linked to their technical names. No model results changed.

Eighth revision, 3 October 2026. Precision of wording, after an author's review. "Years per detection" is now stated as a national figure, across all 175 units, not a wait per offender unit. Recommendation 10 is described as a call to consider a panel, not to create one, and its examples as not naming a statistician. The low base rates are labelled as scenarios, not what the record allows. Rules A–D are distinguished from E, which uses the true expectation; the units bullet no longer claims the two ratios differ by only a few per cent; and "half an offender" becomes an expected 0.5 offender-years. The summary diagram and the threshold-sweep chart carry the national wording. No figures changed.

Ninth revision, 5 October 2026. The base rate is now anchored to the record. A national reference of about one offender-year in 5 million nurse-years (0.1 per 10,000 unit-years) rests on five UK convictions since 1990 over some 25 million nurse-years of published register totals, each case cited, consistent with Forrest (1995) and Gill, Fenton and Lagnado (2022). A neonatal reference of 1 per 10,000 takes the single neonatal case at face value. Rates of 3 to 30 per 10,000 become stress tests. Because the reference is a rate per nurse, risk is now spread in proportion to staff in the main results, with equal risk per unit kept as a check. The code adds the 0.1 scenario, prints each rate per nurse-year, and reports both allocations; the simulated alarm and detection figures are unchanged, and only the arithmetic that combines them with the base rate moved. The headline ratios rise from hundreds or thousands to one, to thousands or tens of thousands to one.

Tenth revision, 5 October 2026. Readability. The summary states its headline in plain words and defines unit-years, nurse-years and offender-years. The five results tables and two figures are numbered and captioned, and the text and the code's output labels refer to them by number. No figures changed.

Eleventh revision, 6 October 2026. Register and sources, after an external review. The text adopts a uniformly formal register and gender-neutral wording for nurses and offenders; terms are used consistently (falsely flagged unit-years in the tables, false-alarm crossings in the threshold sweep); and the summary diagram is redrawn around the own-exposure test. Secondary sources are replaced by primary ones where available, including the Thirlwall recommendations, ONS birth statistics, the RCPCH count of neonatal services and Aylin et al. (Lancet, 2003). No figures changed.

Twelfth revision, 6 October 2026. A correction to offender identification. The rota review resolved ties for the highest attendance in favour of the first nurse on the roster; because the simulated offender was always listed last, ties counted against the offender. Ties are now shared equally among the tied nurses, the expected result of a random tie-break. Identification of an offender in a local unit rises from 23–61% to 27–67%, and the chance that a flagged nurse is the offender moves slightly. Alarm and detection rates are unchanged. The code is version 7.

Thirteenth revision, 7 October 2026. An interactive explorer is added and described. Its default view reproduces every table exactly from the reference run; readers can then vary the assumptions behind each table, with changed settings simulated afresh in the browser by a version of the model checked against the original. The opening now mentions it, and a new section, "Testing the assumptions", describes it before the sources. Figures 1 and 2 carry their numbers in their titles, the threshold-sweep chart's wording is tightened, and the Shipman paragraph now notes that the chart flagged Shipman but could not distinguish him from the 32 other GPs who crossed the threshold. This history is now published as a separate document, linked from the article. No figures changed.

Fourteenth revision, 8 October 2026. The opening now states the article's subject directly: Recommendations 6 and 9 leave no rule for the step from an outcome anomaly to the investigation of a specific person, and the expert review in Recommendation 10 begins only after a person is suspected. The section on how the recommendations fit together now follows the opening directly, so the analytical map introduces the analysis, and a short paragraph opening Step three links it to the two steps before. "In brief" gains three subheadings, its definitions of unit-years, nurse-years and offender-years move to a "Terms used" section near the end, and rule C is tied to Table 1. The headline paragraph under “The scale of the problem” is rewritten as three paragraphs that build from a worked example: a unit of 50 nurses, a chance alarm once in ten years and the conviction-record rate give about 10,000 chance alarms per offender, which the model then refines. The two counting units are defined where they are first used. No figures changed.

Fifteenth revision, 9 October 2026. A check of the text against the code aligned several descriptions with what the model does. The combined row of Table 2 is weighted by reviewed alarm episodes, not by unit numbers, and is now labelled accordingly. The rota review is described as applying to alarm episodes, with consecutive alarmed checks

counted as one and a minimum of two deaths in the window. The text now states the half-death annual floor on the rules A–D baseline, and that annual case-mix variation is drawn from a gamma distribution. Rule E is no longer described as having no false alarms from case mix; it adjusts for case mix, and random variation still triggers it at the set rate. The definition of *q* now states that the same search share applies to every alarm, and Table 5 explains that its smaller columns scale the full-search count of distinct nurses by *q*, which understates them by at most 0.2 a year. No figures changed.

Sixteenth revision, 9 October 2026. A final check of every figure against the model's reference output found no discrepancies. Five descriptions were tightened. The opening example now places the 175 units in England and Wales, as the rest of the article does, rather than the UK. The detection paragraph states that the offender counts as identified when, at the first alarm, the offender is the most-present nurse and passes the own-exposure test, and that each combination of rule, unit type and effect is simulated 10,000 times. A full-time pattern is described as three of the week's fourteen 12-hour shifts, which is what the 21% attendance probability assumes. The limitation on event definitions is now headed "Chance a flagged nurse is the offender" rather than "Posterior events". No figures changed.

Seventeenth revision, 9 October 2026. A note before Table 1 explains that its 2- and 3-standard-deviation lines are shorthand for one-sided Poisson cut-offs at the matching percentiles (97.7% and 99.87%) of the unit's three-year average, which is how the model applies them. The opening example now grounds its chance-alarm rate of once in ten years: it is rule E10's design setting and mid-range for the simpler rules, so the 10,000 chance alarms follow by arithmetic rather than assumption. The example also states that it compares alarms, not people: no one has yet looked at a rota. Both the example and Step one now say that rule E's false-alarm rate is fixed by simulating units with no offender, and that the simpler rules' rates are simulated, not observed. No figures changed.

Eighteenth revision, 9 October 2026. "In brief" gains a short section, "Why a simulation?", before "The scale of the problem". It explains why the historical record cannot measure the error rate of the move from alarm to suspect (direct evidence can settle a single case, but the record scores cases by their outcome, so a wrongful conviction enters it as a success, and it omits chance alarms that led nowhere, nurses quietly cleared and offenders never caught), where the six alarm rules come from (Recommendation 6 leaves the trigger undefined), and, as a numbered list, what the three steps of the analysis will show. The sentence introducing the six rules moves from the opening paragraph into this section. Alarms are now consistently "triggered"; the text had used five verbs (sound, fire, set off, raise, trigger) for one event. No figures changed.

Nineteenth revision, 16 October 2026. A short Disclaimer section is added before the Sources: the analysis is offered in good faith as one interpretation of public material, is not professional advice, makes no finding about any individual or case, and may contain errors. The README carries the same wording. No figures changed.